01. Rank the complexities (slowest → fastest)

Given:

$O(n)$

$O(\log n)$

$O(n^5)$

$O(n \log n)$

$O(1)$

$\therefore\ O(1) < O(\log n) < O(n) < O(n \log n) < O(n^5)$

02 Given the following find Big-O. Complexity

a) $4n^2 + 20n + 9$

soln:

Highest term : n^2

$\therefore\ O(n^2)$

b) $25n^3 + 15$

Highest term $= n^3$

$\therefore\ O(n^3)$

c) $n^3 + 5\log n + 6$

Take n^3 as highest

$\therefore\ O(n^3)$

d) $10^5 + 10^5 n + 10^5 n^2$

Highest term $= n^2$

$\therefore\ O(n^2)$

02 e) $72n(n + 3n^2)$

Soln

$72n(n + 3n^2)$

$72n^2 + 126n^3$

Highest term n^3

$\therefore O(n^3)$

f) $2^8 + 324n - 5$

soln

$2^8 = 256$

Then $256 + 324n - 5$

Highest term n

$\therefore O(n)$

03 O(n) Algorithm to sum 1 to n

```
int Sum to n (int n) {
    int total = 0;
for (int i = 1; 1 <= n; i++) {
    total = total + i;
  }
    return total;
}
```

04. (a) soln

```
X = 4
for i := 0 to ~~do~~ number do
    X := X + 1
endfor
```

(b)

```
choice := get choice ()
  if choice = 1:
 Print "Ready'
  if choice = 2:
for 1:= 0 to cycles do:
  Print i
endfor
```

(c) Two functions.

```
increase (val, iter):
for i: = 1 to iter do:
val:= val + 5
return val

calculate (val, iter)
for i: = 1 to iter do:
val := val + increase (val, iter)
return val
```

04 (d) Calculate (val, iter)

for i := 1 to iter do

val := val + increase (val, i)

return val

05. (i) If $f(n)$ is $O(g(n))$

Soln

$f(n) \leq cg(n)$

$\therefore \leq$

(ii) If $f(n)$ is $\Omega(g(n))$

Soln

$f(n) \geq cg(n)$

$\therefore \geq$

(iii)

06. Show that $4n + 12$ is $O(n)$

soln

let Constant $c = 8$, $k = 3$

for $n \geq 3$

$12 \leq 4n$

Therefore

$4n + 12 \leq 4n + 4n$

$4n + 12 \leq 8n$

Hence

$4n + 12 + O(n)$

$\therefore 4n + 12 \leq 8n \ (n \geq 3)$

07 Show that $5n^2 - 16$ is $O(n^2)$

~~We~~ soln

we must find constant $c > 0$ and $k \geq 0$

Such that $5n^2 - 16 \leq c\,n^2$ for all $n \geq k$

start

$c = 1$

$5n^2 - 16 \leq 1\,n^2 \Rightarrow 4n^2 \leq 16 \Rightarrow n^2 \leq 4 \Rightarrow n \leq 2$

Now choose

$c = 5, \ k = 1$

We need $5n^2 - 16 \leq 5n^2$

Then: $-16 \leq 0$

Since subtracting 16 from $5n^2$

Therefore, with $c = 5$ and $k = 1$:

$5n^2 - 16 \leq 5n^2$ for all $n \geq 1$

$\therefore$ $5n^2 - 16$ is $O(n^2)$